Abstract Algebra — Homework 2

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(a)

By the tower law, $[K : \mathbb{R}] = [K : \mathbb{C}][\mathbb{C} : \mathbb{R}] = 2^{a+1}m$. Let L be the Galois closure of K over $\mathbb{R}$. Set $G = \text{Gal}(L/\mathbb{R})$, so $|G| = [L : \mathbb{R}]$. It's obvious that $|G| = 2^b m'$ for some $b \leq a, m'|m$, m' is odd (also by tower law).

Let $P \leq G$ be a sylow 2-subgroup of G, so $[G : P] = m'$ is odd. Let F be the corresponding intermediate field, we have $[F : \mathbb{R}] = [G : P] = m'$. Suppose $\alpha \in F\backslash\mathbb{R}$, and its minimal irreducible polynomial has degree d ($d \geq 2$). Clearly, $d|[F : \mathbb{R}]$ so d is odd. But any odd-degree polynomial has a real root, which condradicts the irreducibility. Hence, $F = \mathbb{R}$ and $G = P$ is a 2-group.

Therefore, $[L : \mathbb{R}] = 2^b$, by the tower law, $[L : \mathbb{R}] = [L : \mathbb{C}][\mathbb{C} : \mathbb{R}] = 2[L : \mathbb{C}]$, $[L : \mathbb{C}] = 2^{b-1}$. Since $[K : \mathbb{C}]$ divides $[L : \mathbb{C}]$, the only possibility is $a = 0$.

(b)

Assume, $a \leq 1$. Now define $H = \text{Gal}(L/\mathbb{C}) \leq G$, the subgroup is not trivial since $|H| = [L : \mathbb{C}] = [L : K][\mathbb{K} : \mathbb{C}] \leq 2^a \leq 2$

Since 2 is the least prime dividing $|H|$, there exists a normal subgroup of H of index 2, let it be N, $[H : N] = 2$. By the Galois correspondance, the fixed field $E = L^N$ satisfies $[E : \mathbb{C}] = [H : N] = 2$, so E is a degree-2 extension of $\mathbb{C}$.

Pick any $\beta \in E\backslash\mathbb{C}$, its minimal irreducible polynomial over $\mathbb{C}$ must be quadratic:

$$x^2 + bx + c, b, c \in \mathbb{C}$$

By the formula, the roots are $\frac{-b\pm\sqrt{b^2-4ac}}{2}$, now $b^2 - 4ac \in \mathbb{C}$ and every complex number has its square root in $\mathbb{C}$ trivially, so $\beta \in \mathbb{C}$, a contradiction.

Hence, $a = 0$.

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(a)

The minimal polynomial of $\sqrt[3]{2}$ over $\mathbb{Q}$ is $x^3 - 2 = (x - \sqrt[3]{2})(x^2 + 2x + 2)$, which does not split in $\mathbb{Q}(\sqrt[3]{2})$. Hence, $\mathbb{Q}(\sqrt[3]{2})/\mathbb{Q}$ is not a normal extension.

(b)

Since $\mathbb{Q}(\sqrt[4]{2}, \zeta_8)/\mathbb{Q}$ contains all roots of $x^4 - 2$ ($\sqrt[4]{2}, -\sqrt[4]{2}, \zeta_8^2\sqrt[4]{2}, -\zeta_8^2\sqrt[4]{2}$), it's a splitting field for the irreducible polynomial (in $\mathbb{Q}[x]$) $x^4 - 2$. Every splitting field is normal, so $\mathbb{Q}(\sqrt[4]{2}, \zeta_8)/\mathbb{Q}$ is normal.

Clearly, $\mathbb{Q}(\sqrt[4]{2}, \zeta_8) = \mathbb{Q}(\sqrt[4]{2}, i)$. By tower law, $[\mathbb{Q}(\sqrt[4]{2}), i : \mathbb{Q}] = [\mathbb{Q}(\sqrt[4]{2}, i) : \mathbb{Q}(i)][\mathbb{Q}(i) : \mathbb{Q}] = 4 \times 2 = 8$. Hence, the degree for the extension is 8.